

# AMAN KUMAR BHAMBOO

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## SUMMARY

Data analyst who is essentially married to SQL and Tableau. I build robust ETL pipelines that turn massive, fragmented datasets (we're talking 1GB+) into clean, interactive dashboards that even non-technical stakeholders can easily digest. An IEEE-published researcher with a knack for finding the exact signal in the noise, ensuring your business decisions are backed by hard facts, not just gut feelings.

## ACHIEVEMENTS

### Research Publications

"Deep Learning-Based Approaches for Efficient Brain MRI Segmentation: A Comprehensive Analysis," IEEE STCR 2025, pp. 1-9, doi: 10.1109/STCR62650.2025.11020327.

- Aggregated MRI datasets across multiple hospital sources and benchmarked U-Net, ResNet, and transformer-based architectures for brain MRI segmentation across accuracy, computational efficiency, and clinical applicability in resource-constrained environments.

### Open Source Contributions

#### Contributor | GirlScript Summer of Code (GSSoC) 2026

- Contributed to open-source data engineering and python-based repositories, optimizing workflows and resolving critical bugs during the 3-month coding program.

## PROJECTS

### Corporate Revenue Leakage & Discount Audit Dashboard

GitHub | Tableau Dashboard

- **Accomplished multi-dimensional data analysis** of corporate profit erosion across a TPC-H dataset (1.5M orders, 100K customers), processing over **229 Billion in gross transactions** across 27 quarters, by designing 17 complex derived tables within a custom PostgreSQL `derivations` schema.
- **Isolated 11.48 Billion in discount leakage** and quantified pricing anomalies across 200K products and 5 market segments, by conducting a dual-axis "Financial Tug-of-War" demand elasticity trend analysis across 11 distinct discount levels (0–10%).
- **Streamlined corporate financial auditing** and automated margin erosion tracking for executive stakeholders, by building an interactive Tableau dashboard mapping operational leakage flags down to 1,000 individual clerk levels to optimize urgent order discounting.

### Customer Revenue Health

GitHub | Tableau Dashboard

- **Optimized data integrity for 100K e-commerce orders**, eliminating 261K duplicate rows (26% of geolocation data) across 738K records with zero data loss, by architecting a Python, pandas, and SQLAlchemy pipeline to clean and ingest 9 raw CSV files into PostgreSQL.
- **Formulated 6 critical customer health metrics** (CRR, RPR, LTV, Churn, CSAT, NPS), calibrating a realistic +35 NPS over a naive -100 bias, by programming advanced SQL window functions, cohort pivot queries, and text-timestamp casting logic.
- **Identified a critical >70% first-30-day customer churn gap** and low 8% repeat purchase rate, pinpointing regional logistics bottlenecks (**29+ days shipping in Roraima/Amapá vs. 8 days in São Paulo**), by analyzing end-to-end order delivery lifecycles.
- **Translated complex e-commerce behaviors into strategic post-purchase interventions**, designing a 4-page interactive Tableau Public dashboard (cohort heatmaps and geographic performance) to drive a high-ROI 7-day email nudge strategy.

### Mapping Crime Trends in Los Angeles

GitHub | Tableau Dashboard

- **Optimized data integrity for 3.1M+ historical LAPD records**, standardizing 26 disparate metrics and resolving data gaps across 1.4M weapon entries and 1.8M cross-streets, by architecting an end-to-end SQL cleaning pipeline that transformed raw integers into standardized temporal features.
- **Uncovered a massive 68.8% crime surge and critical case resolution disparities**—revealing that 38% of incidents cluster on Friday evenings and property crimes drop to a near-unsolved 1.07% clearance rate—by engineering an interactive multi-page Tableau dashboard.

## EDUCATION

### Vellore Institute of Technology

Bachelor of Technology in Computer Science (*Final Year*, CGPA - 7.8)

Bhopal  
2022–2026

## TECHNICAL SKILLS

**Languages & Databases** Python · SQL · PostgreSQL · MySQL

**Cloud & Engineering** Microsoft Azure · Azure Data Factory · Azure SQL

**Libraries & Analytics** pandas · NumPy · matplotlib · seaborn

**Business Intelligence** Tableau